AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	NTT Chicago CH1, IL -- station KORD, America/Chicago
Committed pair	OSM 1154636644 -> 1446350370 NTT Chicago CH1 / NTT Chicago CH4
Facade gap	79.0 m between the two halls
Weather record	43,775 real hourly records from KORD
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.6652 C at 270 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2022-05-15 (chatter)
Plant limit	21.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 8 of 24 hours with 1 mode change. That is 3 more than the reactive incumbent, at 0 unsafe hours against its 0.

Agent	8 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	5 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	no	dry-bulb	21.532	21.0	21.114	1.058
01	mechanical	no	dry-bulb	21.201	21.0	20.564	0.955
02	mechanical	no	dry-bulb	21.280	21.0	20.574	0.996
03	mechanical	no	dry-bulb	22.116	21.0	21.124	1.035
04	mechanical	no	dry-bulb	21.351	21.0	21.200	0.941
05	mechanical	no	dry-bulb	21.077	21.0	20.692	0.841
06	mechanical	no	dry-bulb	22.055	21.0	20.692	0.876
07	mechanical	no	dry-bulb	22.902	21.0	21.121	1.352
08	mechanical	no	dry-bulb	23.568	21.0	20.560	1.884
09	mechanical	no	dry-bulb	25.231	21.0	22.780	1.925

10	mechanical	no	dry-bulb	25.727	21.0	23.330	1.767
11	mechanical	no	dry-bulb	25.389	21.0	24.440	1.339
12	mechanical	no	dry-bulb	25.272	21.0	21.655	1.545
13	mechanical	no	dry-bulb	24.288	21.0	21.110	1.178
14	mechanical	no	dry-bulb	24.299	21.0	21.111	0.979
15	mechanical	no	dry-bulb	22.694	21.0	22.326	1.115
16	FREE	yes	-	20.757	21.0	18.337	1.215
17	FREE	yes	-	20.699	21.0	18.927	1.060
18	FREE	yes	-	20.773	21.0	18.446	1.093
19	FREE	yes	-	18.359	21.0	17.828	1.304
20	FREE	yes	-	18.605	21.0	17.220	1.335
21	FREE	yes	-	17.491	21.0	16.110	1.186
22	FREE	yes	-	17.307	21.0	16.159	1.047
23	FREE	yes	-	16.970	21.0	15.599	1.194

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 21.531 C against a 21.0 C limit, so it fails by 0.531 C. It would take a limit 0.531 C higher, or a bound 0.531 C tighter, to change this hour.

01:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 21.201 C against a 21.0 C limit, so it fails by 0.201 C. It would take a limit 0.201 C higher, or a bound 0.201 C tighter, to change this hour.

02:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 21.281 C against a 21.0 C limit, so it fails by 0.281 C. It would take a limit 0.281 C higher, or a bound 0.281 C tighter, to change this hour.

03:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 22.116 C against a 21.0 C limit, so it fails by 1.116 C. It would take a limit 1.116 C higher, or a bound 1.116 C tighter, to change this hour.

04:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 21.351 C against a 21.0 C limit, so it fails by 0.351 C. It would take a limit 0.351 C higher, or a bound 0.351 C tighter, to change this hour.

05:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 21.077 C against a 21.0 C limit, so it fails by 0.077 C. It would take a limit 0.077 C higher, or a bound 0.077 C tighter, to change this hour.

06:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 22.054 C against a 21.0 C limit, so it fails by 1.054 C. It would take a limit 1.054 C higher, or a bound 1.054 C tighter, to change this hour.

07:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 22.902 C against a 21.0 C limit, so it fails by 1.902 C. It would take a limit 1.902 C higher, or a bound 1.902 C tighter, to change this hour.

08:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 23.568 C against a 21.0 C limit, so it fails by 2.568 C. It would take a limit 2.568 C higher, or a bound 2.568 C tighter, to change this hour.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.231 C against a 21.0 C limit, so it fails by 4.231 C. It would take a limit 4.231 C higher, or a bound 4.231 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.727 C against a 21.0 C limit, so it fails by 4.727 C. It would take a limit 4.727 C higher, or a bound 4.727 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.389 C against a 21.0 C limit, so it fails by 4.389 C. It would take a limit 4.389 C higher, or a bound 4.389 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.272 C against a 21.0 C limit, so it fails by 4.272 C. It would take a limit 4.272 C higher, or a bound 4.272 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.288 C against a 21.0 C limit, so it fails by 3.288 C. It would take a limit 3.288 C higher, or a bound 3.288 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.298 C against a 21.0 C limit, so it fails by 3.298 C. It would take a limit 3.298 C higher, or a bound 3.298 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.694 C against a 21.0 C limit, so it fails by 1.694 C. It would take a limit 1.694 C higher, or a bound 1.694 C tighter, to change this hour.

16:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 20.757 C, which is 0.243 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.215 C, and the margin is measured, not chosen: 1.008 C of group-conditional forecast error for this hour of day, plus 0.2068 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

17:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 20.699 C, which is 0.301 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.060 C, and the margin is measured, not chosen: 0.830 C of group-conditional forecast error for this hour of day, plus 0.2300 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 20.773 C, which is 0.227 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.093 C, and the margin is measured, not chosen: 0.864 C of group-conditional forecast error for this hour of day, plus 0.2283 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.359 C, which is 2.641 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.304 C, and the margin is measured, not chosen: 1.076 C of group-conditional forecast error for this hour of day, plus 0.2286 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.605 C, which is 2.395 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.335 C, and the margin is measured, not chosen: 1.096 C of group-conditional forecast error for this hour of day, plus 0.2398 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.491 C, which is 3.509 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.186 C, and the margin is measured, not chosen: 0.964 C of group-conditional forecast error for this hour of day, plus 0.2222 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.307 C, which is 3.693 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.047 C, and the margin is measured, not chosen: 0.801 C of group-conditional forecast error for this hour of day, plus 0.2461 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.970 C, which is 4.030 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.194 C, and the margin is measured, not chosen: 0.953 C of group-conditional forecast error for this hour of day, plus 0.2412 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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